Experimental Design: Federated Learning for Tumor Detection

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AI disclaimer: We used AI to ideate, discuss, and create this PDF for the experimental design.

Task

We predict whether a tissue image contains tumor (tumor epithelium) or not, using the PathMNIST dataset. The dataset comes with three separate groups of images:

- **Training (about 90,000 images):** what the models learn from. These are split among 4 mock hospitals.
- **Validation (about 10,000 images):** used to make choices during the experiment, such as the best E and the best round or epoch to keep.
- **Test (about 7,200 images):** used only once, for the final results.

No model trains on validation or test images, and no choices are made using the test images.

Key Terms

- **Centralized:** one model trained on all training data pooled together, with no hospitals or server. This is the reference (ceiling) for performance, but it would not preserve privacy in practice, since every hospital's data would have to be collected in one place.
- **Federated:** each hospital trains its own copy of the model on its own data, and only the weights are shared with the server, which averages them into a global model over several rounds. No hospital's data ever leaves it. The global model from the best round (by validation F1) is what gets evaluated.
- **Single hospital (H0–H3):** one hospital trains a model on only its own data, with no server and no sharing. This is the baseline that federated training is meant to beat.
- **Epoch:** one full pass over a dataset. A **local epoch** is one pass by a hospital over its own data.
- **Round:** one sync with the server. Each hospital sends its weights to the server, the server averages them into global weights, and sends those back to every hospital.
- **E:** the number of local epochs each hospital trains per round. E × R (rounds) is the total number of passes each hospital makes over its own data.
- **Data skew:** how unevenly data is spread across hospitals. We use three splits of the training data:
  - **iid (independent and identically distributed):** the training images are shuffled randomly and divided evenly, so every hospital gets a representative sample: about 22,500 images each, with about 14% tumor, matching the full dataset. This is the no-skew baseline.
  - **label_skew:** every hospital gets the same number of images, but a different share of tumor cases, from about 25% at H0 down to about 2% at H3. This mimics a cancer center versus a small clinic.
  - **size_skew:** every hospital has the same tumor share as the full dataset, but one hospital holds half the training data and the other three split the rest evenly.

Fixed Settings

Setting	Value
Models	Logistic regression (LR); deep neural network (NN) with two hidden layers, with layer sizes kept fixed across all experiments (both from scratch in NumPy)
Batch size	64 (mini-batch)
Training budget	50 epochs (in federated training, 50 local epochs per hospital, so $E \times R = 50$); the best validation checkpoint is kept
Main metric	F1, where an image counts as tumor if the predicted probability is 0.5 or higher (AUC reported alongside)
Aggregation	FedAvg: the server averages the hospitals' weights, with hospitals that have more images counting more
Seeds	3 (seeds 0, 1, and 2) for the final comparison, for both models

Phase 1: Learning Rate (η) Sanity Check

For each model (LR and NN), train centrally with $\eta = 0.01$, mini-batches of 64, for about 10 epochs. Confirm the loss decreases steadily, with no sudden jumps or other misbehaviors. If it misbehaves, try 0.001 or 0.1. We expect the NN to perform at least as well as LR. If it does clearly worse, that points to a bug in the neural network implementation or a poor η. The chosen η for each model is used in all later phases.

Phase 2: Choosing E (Local Epochs)

E is the number of times each hospital trains on its own data before syncing with the server. More E means less communication, but each hospital fits its own data more closely and drifts further from the others, which can hurt the global model when hospitals' data differs.

For each model and each split, run federated training with E = 1, 5, and 10 (50, 10, and 5 rounds, so 50 local epochs per hospital in each case), using seed 0. Check the global model's validation F1 after every round and keep the best. Each split uses its own best E in Phase 3, and the full table is also reported as a finding on how the best E changes with data skew.

Results are recorded in one table per model (validation F1):

Split	E = 1 (50 rounds)	E = 5 (10 rounds)	E = 10 (5 rounds)
iid			
label_skew			
size_skew			

Phase 3: Main Comparison

For each model:

1. **Centralized:** train on all training data. It doesn't depend on the split, so it is trained once per seed and used as the reference (ceiling) for every split.
2. **For each split:** train federated across the 4 hospitals using that split's best E, and train each hospital alone on its own data.

3. Evaluate each model's best checkpoint on the test set, recording F1 and AUC.
4. Repeat with seeds 0, 1, and 2. Fixed seeds make every run reproducible, and tables report the average across the three seeds, which is more reliable than any single run.

Results are recorded in one table per model (test F1):

Split	E used	Centralized	Federated	H0	H1	H2	H3
iid							
label_skew							
size_skew							

Model Comparison (LR vs. NN)

Using the Phase 3 runs (no extra runs needed), we compare the two models to answer one question: is the NN's accuracy gain worth its extra computation and communication? Each model uses its own η and best E. Training time counts computation only, measured on the same machine for both models.

Results are recorded in one table:

Measure	LR	NN
Number of weights		
Size of the weights sent per round		
Training time per epoch (centralized)		
Centralized test F1		
Federated test F1, iid		
Federated test F1, label_skew		
Federated test F1, size_skew		

Success Criteria

Each criterion is checked separately for LR and NN, using the test results.

- **Main:** on the label_skew and size_skew splits, federated F1 is at least 0.02 higher than the average single-hospital F1, for all three seeds. The iid split is also run and reported, but a smaller gap is expected there, since each hospital already has a representative sample of the data.
- **Per hospital:** for each split, count how many hospitals score higher with the federated global model than with the model they trained alone.
- **Reasonable accuracy:** on each split, federated F1 (averaged over the three seeds) is no more than 0.05 below centralized F1.

Write-up

- Federated vs. single-hospital performance across splits, and which hospitals benefit most
- How close federated training gets to centralized training
- E vs. accuracy: how much communication can be reduced, and how this depends on data skew
- LR vs. NN: whether the NN’s F1 gain is worth its extra computation and communication